

Tema 2

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1. Introducerea unei opțiuni de simulare (-contor:LD 0/1) cu ajutorul căreia se poate determina numărul de instrucțiuni cu referire la memorie de tip Load și respectiv Store.

Mai intai , am declarat variabilele

```
/* contor ld / st*/
```

```
1.static unsigned int opt_load;
```

```
/* track number of loads */
```

```
2.static counter_t contor_loads = 0;
```

```
/* track number of stores */
```

```
3.static counter_t contor_stores = 0;
```

Apoi am adaugat optiunea -contor:LD (0 pentru numarul de instructiuni de tipul LOAD , iar 1 pentru numarul de instructiune de tipul STORE)

```
opt_reg_int(odv, "-contor:LD",  
            "1 = numar LOAD-uri, 0 = numar STORE-uri",  
            &opt_load, /* default */0,  
            /* print */TRUE, /* format */NULL);
```

În sim_reg_stats , am adaugat afisarea :

```
if (opt_load == 1)  
{  
    stat_reg_counter(sdb, "contor_loads",  
                    "numar total LOAD-uri",  
                    &contor_loads, contor_loads, NULL);  
}  
else
```

```

{
stat_reg_counter(sdb, "contor_stores",
    "numar total STORE-uri",
    &contor_stores, contor_stores, NULL);
}

```

Iar pe urma in main verific tipul instructiunii si incrementez contorul :

```

if (MD_OP_FLAGS(op) & F_LOAD)
    contor_loads++;

if (MD_OP_FLAGS(op) & F_STORE)
    contor_stores++;

```

Benchmark de test ales : cc1

Linie de comanda : `./sim-LVPred -max:inst 25000000 -contor:LD 0 ../tema1/cc1.ss < ../tema1/1stmt.i` , Output:

```

sim_num_insn      25000000 # total number of instructions executed
valuePrediction    0 # total number of correctly predicted values
classifiedPred     0 # number of loads classified as predictable
classifiedUnpred   0 # number of loads classified as unpredictable
predictable       0 # correctly classified predictable loads
unpredictable     0 # correctly classified unpredictable loads
sim_num_refs      9312069 # total number of loads and stores executed
sim_num_loads     6067498 # total number of loads executed
T1                625682302 # T1
T2                625682302 # T2
sim_num_refs      9312069 # total number of loads and stores executed
sim_elapsed_time   144 # total simulation time in seconds
sim_inst_rate     173611.1111 # simulation speed (in insts/sec)
sim_num_branches  5280405 # total number of branches executed
sim_IPB           4.7345 # instruction per branch
bpred_bimod.lookups 5280405 # total number of bpred lookups
bpred_bimod.updates 5280405 # total number of updates
bpred_bimod.addr_hits 4617208 # total number of address-predicted hits
bpred_bimod.dir_hits 4745858 # total number of direction-predicted hits (includes addr-hits)
bpred_bimod.misses 534547 # total number of misses
bpred_bimod.jr_hits 324391 # total number of address-predicted hits for JR's
bpred_bimod.jr_seen 410892 # total number of JR's seen
bpred_bimod.jr_non_ras_hits.PP 32579 # total number of address-predicted hits for non-RAS JR's
bpred_bimod.jr_non_ras_seen.PP 115776 # total number of non-RAS JR's seen
bpred_bimod.bpred_addr_rate 0.8744 # branch address-prediction rate (i.e., addr-hits/updates)
bpred_bimod.bpred_dir_rate 0.8988 # branch direction-prediction rate (i.e., all-hits/updates)
bpred_bimod.bpred_jr_rate 0.7895 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)
bpred_bimod.bpred_jr_non_ras_rate.PP 0.2814 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/JRs seen)
bpred_bimod.retstack_pushes 295127 # total number of address pushed onto ret-addr stack
bpred_bimod.retstack_pops 295116 # total number of address popped off of ret-addr stack
bpred_bimod.used_ras.PP 295116 # total number of RAS predictions used
bpred_bimod.ras_hits.PP 291812 # total number of RAS hits
bpred_bimod.ras_rate.PP 0.9888 # RAS prediction rate (i.e., RAS hits/used RAS)
contor_stores     3244571 # numar total STORE-uri

```

lar pentru afisarea contorului pentru nr de loads din program , voi folosi linia de comanda :
./sim-LVPred -max:inst 25000000 -contor:LD 1 ../tema1/cc1.ss < ../tema1/1stmt.i

Output :

```
sim: ** simulation statistics **
sim_num_insn          25000000 # total number of instructions executed
valuePrediction        0 # total number of correctly predicted values
classifiedPred         0 # number of loads classified as predictable
classifiedUnpred       0 # number of loads classified as unpredictable
predictable           0 # correctly classified predictable loads
unpredictable         0 # correctly classified unpredictable loads
sim_num_refs          9312069 # total number of loads and stores executed
sim_num_loads         6067498 # total number of loads executed
T1                    625682302 # T1
T2                    625682302 # T2
sim_num_refs          9312069 # total number of loads and stores executed
sim_elapsed_time      153 # total simulation time in seconds
sim_inst_rate        163398.6928 # simulation speed (in insts/sec)
sim_num_branches      5280405 # total number of branches executed
sim_IPB               4.7345 # instruction per branch
bpred_bimod.lookups   5280405 # total number of bpred lookups
bpred_bimod.updates   5280405 # total number of updates
bpred_bimod.addr_hits 4617208 # total number of address-predicted hits
bpred_bimod.dir_hits  4745858 # total number of direction-predicted hits (includes addr-hits)
bpred_bimod.misses    534547 # total number of misses
bpred_bimod.jr_hits   324391 # total number of address-predicted hits for JR's
bpred_bimod.jr_seen   410892 # total number of JR's seen
bpred_bimod.jr_non_ras_hits.PP 32579 # total number of address-predicted hits for non-RAS JR's
bpred_bimod.jr_non_ras_seen.PP 115776 # total number of non-RAS JR's seen
bpred_bimod.bpred_addr_rate 0.8744 # branch address-prediction rate (i.e., addr-hits/updates)
bpred_bimod.bpred_dir_rate  0.8988 # branch direction-prediction rate (i.e., all-hits/updates)
bpred_bimod.bpred_jr_rate   0.7895 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)
bpred_bimod.bpred_jr_non_ras_rate.PP 0.2814 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/JRs seen)
bpred_bimod.retstack_pushes 295127 # total number of address pushed onto ret-addr stack
bpred_bimod.retstack_pops  295116 # total number of address popped off of ret-addr stack
bpred_bimod.used_ras.PP    295116 # total number of RAS predictions used
bpred_bimod.ras_hits.PP    291812 # total number of RAS hits
bpred_bimod.ras_rate.PP    0.9888 # RAS prediction rate (i.e., RAS hits/used RAS)
contor_loads             6067498 # numar total LOAD-uri
```

2. Introducerea unei opțiuni de simulare (-history n) cu ajutorul căreia se pot calcula și afișa primele n numere prime.

Mai intai am adaugat variabila history_n

```
static int history_n = 0 ;
```

Apoi optiunea :

```
opt_reg_int(oddb, "-history_prime",
            "n - primele numere prime",
```

```
&history_n, /* default */0,  
/* print */TRUE, /* format */NULL);
```

In main am apelat functia afiseaza_prime(history_n)

```
if (history_n > 0) {  
    afiseaza_prime(history_n);  
}
```

Functie care afiseaza numerele prime :

```
int is_prime(int x) {  
    if (x < 2) return 0;  
    for (int i = 2; i * i <= x; i++)  
        if (x % i == 0) return 0;  
    return 1;  
}
```

```
void afiseaza_prime(int n) {  
    int count = 0, num = 2;  
  
    printf("Primele %d numere prime:\n", n);  
  
    while (count < n) {  
        if (is_prime(num)) {  
            printf("%d ", num);  
            count++;  
        }  
        num++;  
    }  
}
```

```
printf("\n");
}
```

Linie de comanda : ./sim-LVPred -max:inst 25000000 -contor:LD 1 -history_prime 7
 ../tema1/cc1.ss < ../tema1/1stmt.i

Output:

```
sim: ** starting functional simulation w/ predictors **
Primele 7 numere prime:
2 3 5 7 11 13 17
warning: syscall: sigvec ignored
warning: syscall: sigvec ignored
gcc2_compiled.:
__gnu_compiled_c:
_label_rtx.text
        .align 4
        .proc    0110
_label_rtx:
        !#PROLOGUE# 0
        save %sp,-112,%sp
        !#PROLOGUE# 1
        st %i0,[%fp+68]
        ld [%fp+68],%o0
        ldub [%o0+12],%o1
        and %o1,0xff,%o0
        cmp %o0,40
        be L2
        nop
```

- Introducerea opțiunilor de simulare, a statisticilor, a structurilor de date și funcțiilor aferente în vederea MĂSURĂRII GRADELOR DE LOCALITATE din benchmark-urile SPEC. După fiecare implementare se va scrie într-un document word pașii efectuați, comanda dată și cum arată consola și apoi rezultatul obținut.

În funcția `sim_reg_options` am adăugat următoarele opțiuni de simulare :

- history : câte valori să reținem
- lvpt : dimensiunea tabelii de adrese
- pred : 0 – măsurare localitate 1 – predicție efectivă
- memaddr , 0 – adresă instrucțiune , 1 – adresă cu referință la memorie

Am definit două structuri , și anume : `LVPTvaluelist` , `LVPTaddrlist` .

De exemplu pentru benchmark ul `applu` , am rulat următoarea linie de comandă :

```
./sim-LVPred -max:inst 25000000 -pred 1 -contor:LD 1 -history 1 -lvpt 2048  
../tema1/applu.ss < ../tema1/applu.in
```

Comanda la care am obtinut urmatoarele rezultate :

```
T1 407474026  
Contextual 1  
Taces_predictor 10  
notfoundA 1016  
foundA 1492235  
classifiedPred 1618249  
valuePrediction 1492235  
classifiedUnpred 2244109  
foundAMiss 2370123  
npenload total 252036250  
sim_num_insn 25000000  
sim_num_loads 3863374  
T2 273172876  
Speedup S: 49.163429  
  
sim: ** simulation statistics **  
sim_num_insn          25000000 # total number of instructions executed  
valuePrediction        1492235 # total number of correctly predicted values  
classifiedPred          1618249 # number of loads classified as predictable  
classifiedUnpred        2244109 # number of loads classified as unpredictable  
predictable            1492235 # correctly classified predictable loads  
unpredictable          2086397 # correctly classified unpredictable loads  
sim_num_refs           4555027 # total number of loads and stores executed  
sim_num_loads          3863374 # total number of loads executed  
T1                     407474026 # T1  
T2                     273172876 # T2  
sim_num_refs           4555027 # total number of loads and stores executed  
sim_elapsed_time        1 # total simulation time in seconds  
sim_inst_rate          25000000.0000 # simulation speed (in insts/sec)  
sim_num_branches        647581 # total number of branches executed  
sim_IPB                 38.6052 # instruction per branch  
bpred_bimod.lookups      647581 # total number of bpred lookups  
bpred_bimod.updates      647581 # total number of updates  
bpred_bimod.addr_hits    542858 # total number of address-predicted hits  
bpred_bimod.dir_hits     543247 # total number of direction-predicted hits (includes addr-hits)  
bpred_bimod.misses       104334 # total number of misses  
bpred_bimod.jr_hits       8116 # total number of address-predicted hits for JR's  
bpred_bimod.jr_seen       8156 # total number of JR's seen  
bpred_bimod.jr_non_ras_hits.PP 379 # total number of address-predicted hits for non-RAS JR's  
bpred_bimod.jr_non_ras_seen.PP 411 # total number of non-RAS JR's seen  
bpred_bimod.bpred_addr_rate 0.8383 # branch address-prediction rate (i.e., addr-hits/updates)  
bpred_bimod.bpred_dir_rate 0.8389 # branch direction-prediction rate (i.e., all-hits/updates)  
bpred_bimod.bpred_jr_rate 0.9951 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)  
bpred_bimod.bpred_jr_non_ras_rate.PP 0.9221 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/updates)  
  
bpred_bimod.retstack_pushes 7748 # total number of address pushed onto ret-addr stack  
bpred_bimod.retstack_pops 7745 # total number of address popped off of ret-addr stack  
bpred_bimod.used_ras.PP 7745 # total number of RAS predictions used  
bpred_bimod.ras_hits.PP 7737 # total number of RAS hits  
bpred_bimod.ras_rate.PP 0.9990 # RAS prediction rate (i.e., RAS hits/used RAS)  
contor_loads           3863374 # numar total LOAD-uri
```

Se poate observa ca prin cresterea istoriei , se obtine si o crestere a speed-up-ului si a localitatii valorilor , numarul de predictii corecte a crescut de la 1,49 milioane la 2.40 milioane .

Parametru	history = 1	history = 64
Benchmark	aplu.ss	aplu.ss
History	1	64
LVPT Size	2048	2048
Total Loads	3,863,374	3,863,374
foundA	1492235	2406909
Speed-up	49.16	113.5